A-08 OUTPUT Roll no : 7314

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arinshukla@Arins-MacBook-Air ai % clang++ -std=c++17 assignment-8/robot_path_bfs.cpp -o
a8
arinshukla@Arins-MacBook-Air ai % ./a8
=== BFS Robot Path Planning ===
Enter rows and columns: 4 4
Enter the grid using S, G, X, and O:
SOOX
OXOX
OOOX
XOXG

Initial grid:
S O O X
O X O X
O O O X
X O X G
Robot position: (0, 0)

Step-by-step BFS trace:

Step 1: Exploring the next cell from the front of the queue.
S O O X
O X O X
O O O X
X O X G
Robot position: (0, 0)

Step 2: Discovered (0, 1) and added it to the queue.
S O O X
O X O X
O O O X
X O X G
Robot position: (0, 1)

Step 3: Discovered (1, 0) and added it to the queue.
S O O X
O X O X
O O O X
X O X G
Robot position: (1, 0)
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Step 4: Exploring the next cell from the front of the queue.
S O O X
O X O X
O O O X
X O X G
Robot position: (0, 1)

Step 5: Discovered (0, 2) and added it to the queue.
S O O X
O X O X
O O O X
X O X G
Robot position: (0, 2)

Step 6: Exploring the next cell from the front of the queue.
S O O X
O X O X
O O O X
X O X G
Robot position: (1, 0)

Step 7: Discovered (2, 0) and added it to the queue.
S O O X
O X O X
O O O X
X O X G
Robot position: (2, 0)

Step 8: Exploring the next cell from the front of the queue.
S O O X
O X O X
O O O X
X O X G
Robot position: (0, 2)

Step 9: Discovered (1, 2) and added it to the queue.
S O O X
O X O X
O O O X
X O X G
Robot position: (1, 2)

Step 10: Exploring the next cell from the front of the queue.
S O O X
O X O X
O O O X
X O X G
Robot position: (2, 0)

Step 11: Discovered (2, 1) and added it to the queue.
S O O X
O X O X
O O O X
X O X G
Robot position: (2, 1)

Step 12: Exploring the next cell from the front of the queue.
S O O X
O X O X
O O O X
X O X G
Robot position: (1, 2)

Step 13: Discovered (2, 2) and added it to the queue.
S O O X
O X O X
O O O X
X O X G
Robot position: (2, 2)

Step 14: Exploring the next cell from the front of the queue.
S O O X
O X O X
O O O X
X O X G
Robot position: (2, 1)

Step 15: Discovered (3, 1) and added it to the queue.
S O O X
O X O X
O O O X
X O X G
Robot position: (3, 1)

Step 16: Exploring the next cell from the front of the queue.
S O O X
O X O X
O O O X
X O X G
Robot position: (2, 2)

Step 17: Exploring the next cell from the front of the queue.
S O O X
O X O X
O O O X
X O X G
Robot position: (3, 1)

No path exists from S to G.
arinshukla@Arins-MacBook-Air ai %